Ethical Intelligence

AI/ML Syllabus

Module (with Duration)	Sub-Module / Topics
Introduction to Programming – (4 weeks)	
• Python – Basic to Advanced	• Variables and Data Types. • Conditional Statements. • Loops (for and while). • List, Tuple, Dictionary, Sets. • Functions. • Lambda, map, filter, reduce. • Comprehensions. • Specialized Sorting. • File Handling. • Exception Handling. • Regex. • Iterators and Generators. • Recursion. • OOP
Data Analytics and Visualization – (5 weeks)	
1) Python Libraries	• Numpy • Pandas • Matplotlib • Seaborn • Scikit-Learn • Scipy.stats
2) SQL	• Introduction to DBMS and SQL • SQL Data types and Constraints • DDL and DML Commands(Create, • Select, Update, Delete, Alter, Insert, Drop, Truncate) • Aggregate Functions • Where and Having Clause • Group by • Order by • Union and Joins • Subquery • Inline Calculations • Case-When statements • Datetime Functions • Window Functions • Stored Procedures
3) Tableau	• Basic Charts and Operations • Data Structuring Options • Filter and Calculations • Table Calculation and Analytics • Operations on Dataset

	• Advanced Charts
Data Handling and Preparation – (2 weeks)	
1) Data Preprocessing	• Data Cleaning (handling missing values, outliers, duplicates) • Data Transformation (scaling, normalization, encoding) • Data Wrangling (reshaping, merging, filtering)
2) Exploratory Data Analysis (EDA)	• Univariate and Multivariate Analysis • Visualization • Correlation Matrix
3) Feature Engineering	• Feature Creation • Transformation • Feature Correlation • Selection and Reduction
Machine Learning – (5 weeks)	
1) Mathematics (Applied Maths)	• Linear Algebra • Vectors • Calculus • Probability • Statistics • Hypothesis Testing
2) Supervised ML	• Linear Regression (Polynomial Regression and Regularization) • Gradient Descent Algorithm • Overfitting and Underfitting • Logistic Regression (Confusion Matrix, AU-ROC and PR Curves) • K-Nearest Neighbors • Decision Trees • Ensemble Learning – Bagging, Boosting, Stacking and Cascading • Naïve Bayes • SVM
3) Unsupervised ML	• PCA, t-SNE • Clustering • DB-Scan • Anomaly Detection • Time Series and Forecasting • Recommendation Systems
Deep Learning – (6 weeks)	
1) NeuralNetworks- ANN	• Multilayer Perceptron • Activation Functions (Softmax, tanh, ReLU, Leaky ReLU) • Forward and Back Propagation • TensorFlow and Keras • Optimizers(Gradient Descent with Momentum) • Hyperparameter Tuning for NN • Autoencoders
2) Computer Vision - CNN	• Introduction to CNN • Building CNN Models • Transfer Learning • Object Localization and Detection • Open CV

3) Recurrent NeuralNetworks- RNN	• Sequence Modeling • Fundamentals of RNNs • Backpropagation Through Time (BPTT) • Vanishing/Exploding Gradients • Variants of RNNs(LSTM, GRU) • Bi-directional RNNs • Time Series Forecasting with RNNs
4) Natural Language Processing	• Text Representation and Preprocessing (Tokenization, Stemming, Lemmatization) • Word embeddings – Bag of Words, TF-IDF, Word2Vec, Glove Vector • Named Entity Recognition • NLP Libraries (NLTK, SpaCy) • NLP using LSTM • Attention Mechanism • Encoder-Decoder Architecture • Transformers • BERT Model • GPT Overview • What is Hugging Face and the Transformers library? • Using Pre-Trained Models
Modern AI Stack – (3 weeks)	
1) Generative AI	• Intro to GenAI • What are Large Language Models (LLMs)? • Applications of LLMs • Accessing LLMs using API Key (Gemini & Groq) • Fine-Tuning LLMs – LoRA & QLoRA
2) LLM Framework – LangChain	• Introduction to LangChain • Core Components of LangChain – LLMs, Prompt Templates, Chains, Agents, Tools
3) AI Agents	• What is an AI Agent? • Components of an Agent – Planning, Tool Selection, Memory and Reflection • Building AI Agents
4) Vector DB and RAG	• Introduction to Vector DB • Overview of popular vector DBs (FAISS, Pinecone, etc.) • Storing andQuerying Document Embeddings • Applying Similarity Search • What is RAG? • Differences between RAG search and LLM output • Building a RAG Pipeline using – LangChain, Vector DBs and LLMs
Application and Deployment – (3 days)	
1) Frontend	• Streamlit
2) Version Control System	• Git/GitHub, CI/CD Pipelines
3) Containerization	• Docker
Tech Stack Add-ons – (1 week)	
1) Data Engineering Basics	• What is Data and ETL Pipeline • ETL vs. ELT • Data Sources and Formats • Building a basic ETL Pipeline • Distributed Data Processing (Spark)

	• Data Warehousing Systems
2) Application Framework	• FastAPI
3) API Testing Tool	• Postman